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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester (CBCS) Degree Examinations, NOV. - DEC. 2025**25CHY103-MATERIALS CHEMISTRY AND ENERGY APPLICATIONS**

(For MEC / AERO)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. A working concentration cell is made of two half-cells having
A) Identical electrolytes and identical concentrations B) Identical electrolytes but different concentrations
C) Different electrolytes and identical concentrations D) Different electrolytes and different concentrations
2. The glass electrode is sensitive to ions.
A) Sodium B) Hydrogen
C) Chloride D) Hydroxide
3. The process during charging of a Li-ion battery involves
A) Li^+ ions moving from the cathode to the anode B) Li^+ ions moving from anode to cathode
C) H^+ moving from the cathode to the anode D) H^+ moving from anode to cathode
4. The most commonly used semiconductor material in commercial solar cells is
A) Gallium arsenide (GaAs) B) Germanium (Ge)
C) Cadmium sulfide (CdS) D) Silicon (Si)
5. The accelerated corrosion of iron under defective tin coatings is due to
A) Change in electrode potential B) Acidic environment
C) Small cathodic surface area D) Small anodic surface area
6. Chemical conversion coating is a type of coating.
A) Organic B) Polymer
C) Inorganic D) Alloy
7. In the electroplating process, generally the object to be plated is made as
A) Cathode B) Electrolyte
C) Anode D) Power house
8. In electronics, the abbreviation PCB stands for
A) Polished Circuit Board B) Printed Circular Board
C) Polymer Circuit Board D) Printed Circuit Board
9. Natural rubber is vulcanized. This is to
A) Increase softness B) Improve solubility
C) Reduce molecular weight D) Improve elasticity and strength
10. Which of the following is not a type of moulding process?
A) Injection B) Annealing
C) Extrusion D) Compression
11. Carbon fibre is obtained from heating
A) Polyacrylonitrile B) Polyvinyl alcohol
C) Polymethylmethacrylate D) Polyvinylchloride

12. CVD stands for
 A) Carbon Vapor Dissolution B) Chemical Vapor Deposition
 C) Carbon Vapor Deposition D) Chemical Vapor Dissolution
13. Stainless steel is primarily an alloy of
 A) Iron and Chromium B) Aluminium and Copper
 C) Copper and Nickel D) Zinc and Silver
14. Non-ferrous metals lack as a major constituent in their composition.
 A) Aluminum B) Carbon
 C) Iron D) Copper
15. Alnico alloys are known for their excellent properties
 A) Chemical B) Mechanical
 C) Thermal D) Magnetic
16. Pb extracted from its ores always contains small amounts of and it is known as 'argentiferous lead'.
 A) Ag (Silver) B) Cu (Copper)
 C) Zn (Zinc) D) Fe (Iron)
17. The calorific value of a fuel is measured in
 A) kJ/mol B) kJ/kg
 C) kcal/mol D) kJ/mole
18. Bomb Calorimeter is used to determine of a solid or liquid fuel.
 A) Only GCV B) Only NCV
 C) Both GCV and NCV D) Neither GCV nor NCV
19. Knocking in a diesel engine occurs due to
 A) Early ignition B) Spark plug failure
 C) Delayed ignition D) Excess oxygen
20. Power alcohols contain
 A) 50-60% of petrol B) 70-80% of petrol
 C) 50-60% of ethanol D) 70-80% of ethanol

PART - B: DESCRIPTIVE ANSWER QUESTIONS

80 Marks

Unit – I		Marks	BT*	CO*	PO*
1. a)	With a neat and labelled diagram, explain the construction and working of the calomel electrode. Mention any two advantages of it.	6	2	1	1
b)	Discuss the construction and working of the Li-ion battery.	6	2	1	1
c)	A Concentration cell was constructed by immersing two nickel electrodes in 0.08 M and 0.8 M NiSO_4 solutions. Write the cell scheme and calculate the EMF of the cell at 298K.	4	3	1	1
2. a)	Define ion-selective electrode. Explain the determination of pH using a glass electrode.	6	2	1	1
b)	Draw a methanol-oxygen fuel cell. Explain how it is used to generate electricity. Write any two applications of it.	7	3	1	1
c)	Define photovoltaic cells. Write any two advantages and disadvantages of it.	3	2	1	1
Unit – II					
3. a)	Apply the electrochemical theory to explain corrosion by taking iron as an example.	8	3	2	1
b)	A piece of corroded metal alloy plate was found in a submerged ocean vessel. It was estimated that the original area of the plate was 180 cm^2 and that approximately 3.8 Kg had corroded away during the submersion. Assuming a corrosion penetration rate of 4.9 mm/year for this alloy in seawater, estimate the time of submersion in years. The density of the alloy is 4.5 g/cm^3 .	4	3	2	1
c)	List out any four differences between electroplating and electroless plating.	4	2	2	1

4. a)	Define electroless plating. Apply the concept of electroless plating of copper in the manufacturing of a double-sided printed circuit board (PCB).	8	3	2	1
b)	Write a note on the factors below that affect the rate of corrosion. (i) Relative areas of the anodic and cathodic parts (ii) pH of the medium	4	2	2	1
c)	Discuss the steps involved in the galvanization process.	4	2	2	1

Unit – III

5. a)	In a sample of a polymer, 27 molecules have a molecular mass of 16900g/mol, 19 molecules have a molecular mass of 33740 g/mol, and 39 molecules have a molecular mass of 26480 g/mol. Calculate the number average and weight average molecular mass of the polymer	6	3	3	1
b)	Explain the synthesis properties and applications of Carbon fibre.	6	2	3	1
c)	Discuss the synthesis of carbon nanotubes using the CVD method.	4	2	3	1
6. a)	Discuss the classification of nanomaterials with an example.	6	2	3	1
b)	What is Glass transition temperature (T _g). Analyze any 4 factors that affect the glass transition temperature	7	4	3	1
c)	Explain the preparation of silicon rubber.	3	2	3	1

Unit – IV

7. a)	Define an alloy. Explain any two properties and applications of ferrous metals.	5	2	4	1
b)	Define Brass. Write the properties and applications of brass.	7	2	4	1
c)	Explain the separation of silver from argentiferous lead by Pattinson's process.	4	2	4	1
8. a)	Discuss the phase diagram of the one-component system water in detail.	6	2	4	1
b)	Explain the properties and applications of (i) Magnesium (ii) Duralumin.	8	2	4	1
c)	Write the Gibbs phase rule equation and define the terms used in it.	2	2	4	1

Unit – V

9. a)	Draw an experimental setup and explain the determination of the calorific value of a solid fuel using the Bomb calorimetric method	8	5	5	1
b)	Write any three reactions for the reforming of a petroleum fraction.	6	2	5	1
c)	Mention any two advantages of Power alcohol.	2	2	5	1
10. a)	Define knocking. Explain the mechanism of knocking in a gasoline engine.	8	2	5	1
b)	On burning 0.095 kg of a solid fuel in a bomb calorimeter, the temperature of 2.8 kg of water increases from 25 °C to 28.7 °C. The water equivalent of the calorimeter and the latent heat of steam are 0.385 kg and 2454 kJ/kg, respectively. If fuel contains 0.9 % hydrogen, calculate the gross and net calorific value of the sample. The specific heat of water is 4.184 kJ/kg/ °C.	6	3	5	1
c)	Mention the main purpose of the cracking process.	2	2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
